



1. What is Statistics?

Statistics is the science of **collecting**, **organizing**, **analyzing**, and **interpreting** data.

In real life, data is everywhere — food delivery times, user ratings, stock prices, exam scores, etc.



Why Use Statistics?

- To **understand what's going on**
 - To **make decisions or predictions** based on data
-



Real-Life Example: Swiggy

Swiggy wants to analyze delivery data:

- What's the **average delivery time**?
- What food is **most ordered**?
- Will the **new delivery policy** reduce time?

All these can be answered using **statistics**.



2. Types of Statistics

Type	Purpose	Example
Descriptive	Describe and summarize data	"Average rating is 4.5 stars"
Inferential	Predict from a sample	"Estimate: 70% of users will reorder"

✓ A. Descriptive Statistics

Deals with **describing data** using:

- **Mean** (average)
- **Median** (middle)
- **Mode** (most frequent)
- **Range, Variance, Standard Deviation**
- **Charts** (bar, pie, histogram)

🔧 Example:

"The average order value is ₹320."

✓ B. Inferential Statistics

Uses a **sample** to make predictions about the **whole population**:

- Sampling
- Probability
- Hypothesis testing
- Confidence intervals
- Regression

🔧 Example:

"60% of 500 users liked the new feature — so we estimate 60% of all users will like it."



3. Population vs Sample

Population:

The entire group you want to study.

 Example: *All 1 million Swiggy users in India*

Sample:

A smaller group taken from the population.

 Example: *500 users selected randomly*

We use samples because studying the whole population is **costly and time-consuming**.

4. Measures of Central Tendency

Used to **summarize data** by identifying the center or typical value.

Measure	Meaning	Use Example
Mean	Average value	“Average delivery time: 30 mins”
Median	Middle value	Good when data has outliers
Mode	Most frequent value	“Most common food: Pizza”

Example Dataset:

[25, 30, 30, 35, 50]

- **Mean** = $(25 + 30 + 30 + 35 + 50) / 5 = 34$
 - **Median** = 30
 - **Mode** = 30
-

Why Do We Need Mean, Median, and Mode?

They help answer:

“What’s the most typical value?”

Use:

- **Mean** → Average (use when data is uniform)
 - **Median** → Middle (use when data has outliers)
 - **Mode** → Most frequent (use for categorical/trending data)
-



Example Comparison:

Dataset: [25, 30, 30, 35, 100]

- **Mean** = 44 (influenced by outlier 100)
- **Median** = 30
- **Mode** = 30

👉 Median is better here due to the outlier.



Measures of Dispersion

These tell us how **spread out** the data is — how much it deviates from the center.



Why Important?

Two datasets can have the same average but different spreads:

- **Class A:** [70, 70, 70, 70, 70] → consistent
 - **Class B:** [40, 55, 70, 85, 100] → widely spread
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Types of Dispersion

Measure	Tells You	Notes
Range	Difference between min & max	Simple
Variance	Avg squared distance from mean	Important
Standard Deviation	Square root of variance	Most used
Interquartile Range	Spread of middle 50% values	Useful for box plots

What is Variance?

Variance tells how far each value is from the **mean**.

- **Small Variance** → values close to mean
 - **Large Variance** → values are more spread out
-

Types of Variance

Type	Formula	Use When
Population (σ^2)	$\sigma^2 = (1/N) \sum (x_i - \mu)^2$	You have all population data
Sample (s^2)	$s^2 = (1/n-1) \sum (x_i - \bar{x})^2$	You have a sample , not full data

Population Variance Example

Data: [2, 4, 6, 8]

Mean (μ) = 5

Steps:

- $(2-5)^2 = 9$
- $(4-5)^2 = 1$
- $(6-5)^2 = 1$
- $(8-5)^2 = 9$

$$\sigma^2 = (9 + 1 + 1 + 9) / 4 = 5$$

2 Sample Variance Example

Sample: [2, 4, 6]

$$\text{Mean } (\bar{x}) = (2 + 4 + 6) / 3 = 4$$

Steps:

- $(2-4)^2 = 4$
- $(4-4)^2 = 0$
- $(6-4)^2 = 4$

$$s^2 = (4 + 0 + 4) / (3-1) = 4$$

Why Divide by (n - 1) in Sample Variance?

Key Reason:

When calculating variance from a **sample**, dividing by **(n - 1)** instead of **n** helps avoid **underestimating the true population variance**.

This correction is called **Bessel's Correction**.



Real-World Analogy:

You want to check how consistent your delivery team is across **1,000 deliveries**. But you only check **5** of them.

If you divide by **5 (n)**, you assume the **sample mean** is exactly the **population mean**, which is rarely true.

So, we divide by **(n - 1)** to **correct the bias** — making the estimate more honest and accurate.



A Simple Example:



Population:

[2, 4, 6, 8, 10]

- Population Mean (μ) = $(2 + 4 + 6 + 8 + 10) / 5 = 6$
 - Population Variance (σ^2) =
 $= [(2-6)^2 + (4-6)^2 + (6-6)^2 + (8-6)^2 + (10-6)^2] / 5$
 $= (16 + 4 + 0 + 4 + 16) / 5 = 40 / 5 = 8$
-



Sample:

Take a sample of 3 numbers: [2, 4, 6]

Sample Mean ($\bar{x}$) = $(2 + 4 + 6) / 3 = 4$

Case 1: Divide by n (3)

Variance = $[(2-4)^2 + (4-4)^2 + (6-4)^2] / 3$
 $= (4 + 0 + 4) / 3 = 2.67$

👉 Too **low** compared to the actual population variance (**8**)

Case 2: Divide by (n - 1 = 2)

$$\begin{aligned}\text{Variance} &= [(2-4)^2 + (4-4)^2 + (6-4)^2] / 2 \\ &= (4 + 0 + 4) / 2 = 4\end{aligned}$$

👉 **Closer** to the true variance (8) — still not exact, but more accurate **on average across samples**



The Intuition

When using a **sample**:

- You're **estimating** the mean from the same data.
- This causes the values to appear **less spread out** than they really are.
- So, dividing by **(n - 1)** slightly **boosts** the variance estimate.

This makes the **sample variance an unbiased estimate** of the **true population variance**.



Summary Table

Term	Divided by	Used When
Population Variance	n	You have all data
Sample Variance	n - 1	You have a sample

🧠 The Real Reason: Degrees of Freedom

📌 What are "Degrees of Freedom"?

When calculating sample variance, we divide by $(n - 1)$ instead of n .

Why? Because when we calculate the sample mean, we use up 1 piece of information.

So, out of n values, only $(n - 1)$ are "free to vary" — this is what we call **degrees of freedom**.

Example: Sample of 3 Numbers

Let's say your sample has 3 numbers:

- $x_1 = 2$
- $x_2 = 4$
- $x_3 = ?$ (unknown for now)
- Sample mean = 5

Using the Mean Formula:

$$(x_1 + x_2 + x_3) / 3 = 5$$

$$(2 + 4 + x_3) / 3 = 5$$

$$(6 + x_3) / 3 = 5$$

$$6 + x_3 = 15$$

$$x_3 = 9$$

What Just Happened?

As soon as we fix two values (x_1 and x_2) and the sample mean, the third value (x_3) is forced to be a specific number.

So even though we have 3 values, only 2 are truly independent. That means we only have **$n - 1 = 2$ degrees of freedom**.

Why It Matters in Variance

- Variance measures how much values differ from the mean
- But since we **used the mean** from the data, one value is **not independent**
- So we adjust by dividing by **$(n - 1)$** instead of **n**

What is Standard Deviation?

Standard Deviation is a measure of how spread out the numbers are from the mean (average) of the data.

It answers:

👉 "On average, how far is each data point from the mean?"

Why Use Standard Deviation?

- It tells you how consistent or varied your data is.
 - Used in finance, data science, machine learning, quality control, etc.
-

If Standard Deviation is:

- Small → Data points are close to the mean (less spread)
 - Large → Data points are far from the mean (more spread)
-

Formula

For **Population Standard Deviation (σ)**:

$$\sigma = \sqrt{\frac{(x_1 - \mu)^2 + (x_2 - \mu)^2 + \dots + (x_N - \mu)^2}{N}}$$

For **Sample Standard Deviation (s)**:

$$s = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{(n - 1)}}$$

Example (Step-by-Step)

Sample Data: [2, 4, 6, 8]

Step 1: Find the mean

$$\text{Mean } (\bar{x}) = (2 + 4 + 6 + 8) / 4 = 20 / 4 = 5$$

Step 2: Subtract the mean from each value and square it

$$(2 - 5)^2 = 9$$

$$(4 - 5)^2 = 1$$

$$(6 - 5)^2 = 1$$

$$(8 - 5)^2 = 9$$

Step 3: Add the squared differences

$$9 + 1 + 1 + 9 = 20$$

Step 4: Divide by (n - 1)

$$n = 4 \rightarrow n - 1 = 3$$

$$\text{Variance } (s^2) = 20 / 3 \approx 6.67$$

Step 5: Take the square root

$$\text{Standard Deviation } (s) = \sqrt{6.67} \approx 2.58$$

✓ This is the **sample standard deviation**.

Real-Life Examples

Example	What SD Tells
Stock returns	High SD = risky, Low SD = stable
Exam scores	Low SD = consistent scores
Delivery times	High SD = inconsistent timing

Relation to Variance

Standard Deviation = Square Root of Variance

- Both measure spread
 - Standard deviation is in original units(variance is in squared unit) → easier to interpret
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Summary

Term	Description	Formula
Variance	Avg. squared distance from mean	$\text{sum}(x - \bar{x})^2 / (n - 1)$
Standard Deviation	Square root of variance	$\sqrt{\text{Variance}}$

What is a Variable?

A **variable** is anything that can **take different values**.

It's a characteristic, number, or quantity that can **change or vary** from one data point to another.

Examples:

- Height of students (could be 160 cm, 170 cm, etc.)
 - Temperature of a city (could be 25°C, 30°C, etc.)
 - Age of people (20, 25, 30 years)
 - Color of a car (Red, Blue, Black)
-

Types of Variables

Variables are mainly divided into two broad categories:

1 Qualitative (Categorical) Variables

- **Describe categories or groups** — they tell you "what kind" or "which group."
- Values are **labels or names**, not numbers you calculate with.

Examples:

- Gender: Male, Female, Other
- Car color: Red, Blue, Green
- Education level: High School, Bachelor's, Master's

Types of Qualitative Variables:

- **Nominal:** Categories with no order
Example: Eye color (Blue, Brown, Green)
 - **Ordinal:** Categories with a clear order
Example: Customer satisfaction (Poor, Fair, Good, Excellent)
-

2 Quantitative (Numerical) Variables

- **Represent measurable quantities** — numbers you can do math on.

Types of Quantitative Variables:

- **Discrete Variables**
 - Take specific values (often whole numbers)
 - Example: Number of cars, number of students
 - **Continuous Variables**
 - Can take any value within a range (including decimals)
 - Example: Height (160.5 cm), weight (70.2 kg), temperature (25.3°C)
-

Summary Table

Variable Type	Description	Examples
Qualitative	Categories or labels	Gender, Car color, Satisfaction
- Nominal	No order	Eye color, Marital status
- Ordinal	Ordered categories	Rating scale, Education level
Quantitative	Numerical values	Age, Weight, Temperature
- Discrete	Countable numbers	Number of students, Cars

- Continuous Any value in a range Height, Time, Temperature

What is a Random Variable?

A **random variable** is a variable that takes on different **numerical values** based on the outcome of a **random experiment**.

It's like a **bridge between probability and numbers**.

Real-World Example:

- Toss a coin:
 - Heads = 1
 - Tails = 0
 - The outcome is random, but you're assigning a **number** to it.
 - Roll a die:
 - Outcomes: 1, 2, 3, 4, 5, 6
 - Each outcome is random, and the result is your **random variable**.
-

Definition:

A **random variable** is a function that assigns a **numeric value** to each outcome in a sample space of a **random experiment**.

Types of Random Variables

There are two main types:

1 Discrete Random Variable

- Takes on a **finite or countable** number of values.
- You can **list all possible outcomes**.
- Usually comes from **counting**.

✓ Examples:

- Number of heads in 3 coin tosses: 0, 1, 2, 3
- Number of students in a class: 20, 21, 22, ...
- Rolling a die: values from 1 to 6



Represented using a probability mass function (PMF)

2 Continuous Random Variable

- Takes on **infinite values** within a range.
- Comes from **measuring**, not counting.
- You **cannot list all values** — decimals included.

✓ Examples:

- Height of people (like 172.3 cm, 180.5 cm)
- Time taken to run a race (12.6s, 13.01s, ...)
- Temperature in a city (25.5°C, 27.3°C, etc.)



Represented using a probability density function (PDF)

Key Differences:

Feature	Discrete Random Variable	Continuous Random Variable
Takes values from	Countable set (finite or infinite)	Uncountable (real numbers in a range)
Example	No. of heads in coin toss	Height, Weight, Temperature
Uses	PMF (Probability Mass Function)	PDF (Probability Density Function)
Can list values?	✓ Yes	✗ No

Normal Variable vs Random Variable

Feature	Normal Variable	Random Variable
◆ Definition	A variable that holds a known value	A variable whose value is determined by chance
 Value	Fixed or assigned directly by you	Comes from a random experiment (uncertain outcome)
 Based on Probability?	✗ No — it's just data	✓ Yes — linked to probability
 Examples	- Age = 25 - Salary = ₹50,000	- X = Number of heads in 3 coin tosses - Y = Time taken to solve a CAPTCHA
 Use in Statistics	Used in descriptive statistics	Used in probability and inferential statistics



What is a Histogram?

A **histogram** is a type of bar graph that shows the **distribution of numerical data**. It helps you see **how frequently** different ranges of values (called "bins") occur in a dataset.



Think of it like this:

Imagine you measured the ages of 20 people, and you grouped them like this:

Age Range (Bin)	Number of People
10–19	2
20–29	5
30–39	8
40–49	4
50–59	1

A **histogram** would show **bars** for each age group with heights representing the **count** of people in that range.



Key Terms:

Term	Meaning
Bins	Intervals or ranges of values (like 0–10, 10–20, etc.)
Frequency	How many values fall into each bin
Height of bar	Represents frequency — taller bars mean more values in that range



When to Use a Histogram?

Use it when:

- You want to understand how data is **spread or distributed**
 - You're dealing with **continuous numeric data** (like height, marks, salary, etc.)
-

Real-World Example:

Suppose you collect exam scores of 100 students (out of 100). You can group the scores like this:

Score Range	Number of Students
0–10	1
11–20	3
21–30	5
...	...
91–100	12

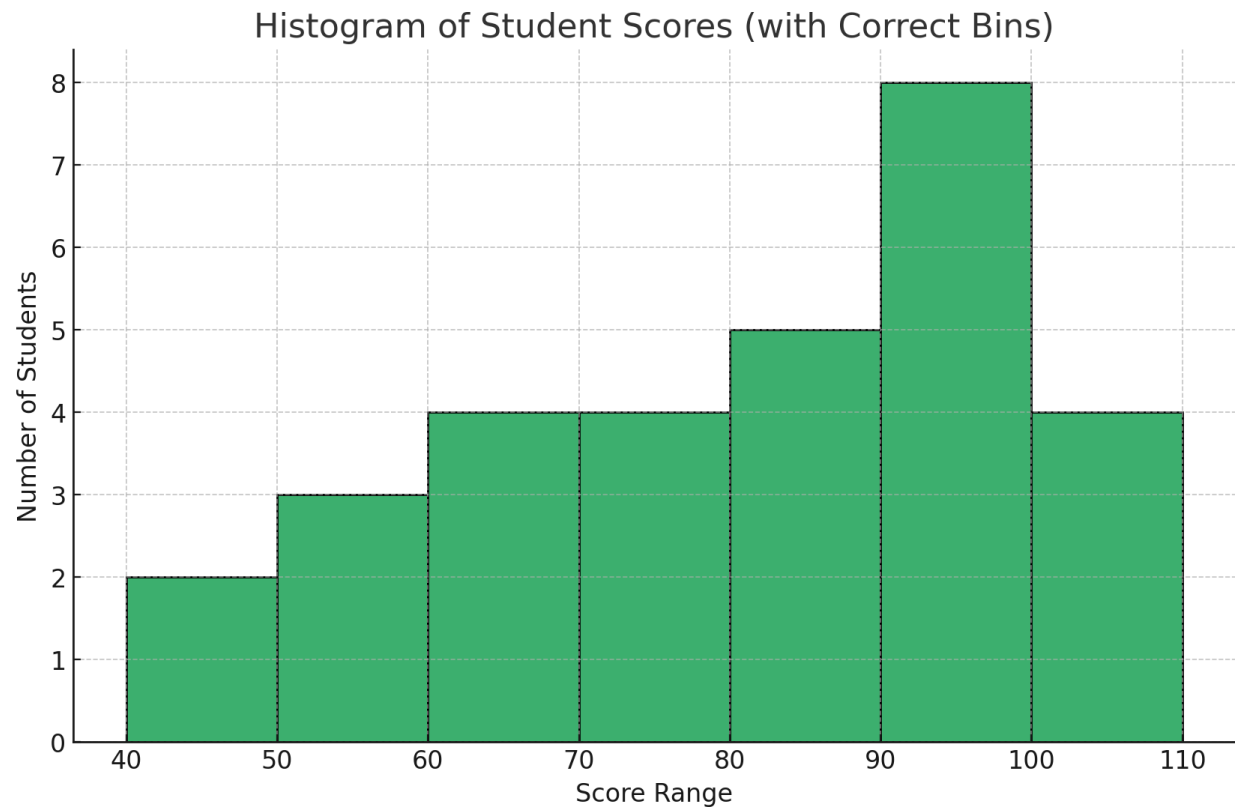
The histogram will visually show you:

- Where most students scored
 - If the scores are clustered or spread out
 - Whether the distribution is symmetric or skewed
-

Histogram vs Bar Chart

Feature	Histogram	Bar Chart
Type of data	Numeric (continuous)	Categorical

Bars touch?	✓ Yes (no gap between bins)	✗ No (bars are separate)
Purpose	Show distribution of values	Compare categories



✓ 1. Percentage

📘 Definition:

A **percentage** is a number or ratio expressed as a fraction of 100.

📖 Formula:

$$\text{Percentage} = (\text{Part} / \text{Whole}) \times 100$$

🎯 Example:

You scored 45 out of 50 in a test.

$$\text{Percentage} = (45 / 50) \times 100 = \mathbf{90\%}$$



2. Percentile



Definition:

A **percentile** indicates the value below which a given percentage of observations in a group fall. For example, the **90th percentile** means 90% of values are below that number.

Percentiles divide the dataset into 100 parts.



Example:

In a test of 100 students, if you are in the **80th percentile**, it means:

- You scored better than 80 students.
- Only 20 students scored better than you.



Use Case:

Used in standardized tests, salary comparisons, data ranking, etc.



3. Quartile



Definition:

Quartiles divide a dataset into four equal parts.

Quartile	What It Represents
Q1	25% of the data is below this value
Q2	50% of the data is below (median)
Q3	75% of the data is below this value



Example:

Scores (sorted): 10, 20, 30, 40, 50, 60, 70, 80

- $Q1 = 25\text{th percentile} = 25$
- $Q2 = \text{Median} = (40 + 50)/2 = 45$
- $Q3 = 75\text{th percentile} = 65$



Comparison Table

Concept	Division	Example Meaning
Percentage	Out of 100	75% marks → 75 out of 100
Percentile	Rank Position	85th percentile → better than 85% of the group
Quartile	4 parts	$Q1 = 25\text{th percentile}$, $Q2 = 50\text{th (Median)}$, $Q3 = 75\text{th}$



Percentile Formulas

1. Percentile Position Formula (to find value at a given percentile):

$$\text{Position} = (P / 100) \times (n + 1)$$

Where:

- P = desired percentile (e.g., 25, 50, 70)
- n = number of values in the dataset

👉 If result is decimal → interpolate between values at floor and ceiling positions.

♦ Example:

Find the **70th percentile** in a dataset of 11 sorted numbers:

$$\text{Position} = (70 / 100) \times (11 + 1) = 0.7 \times 12 = \mathbf{8.4}$$

Look between the 8th and 9th values → interpolate.

2. Percentile Rank Formula (to find rank of a given value):

Percentile Rank = (Number of values less than x / n) $\times$ 100

Where:

- x = value in question
- n = total number of values

♦ Example:

Data = [10, 20, 30, 40, 50]

Find percentile rank of **30**:

Percentile Rank = $(2 / 5) \times 100 = 40$

So, 30 is at the 40th percentile.

Practice Questions

Q1: Find the 50th percentile (P50)

Data (sorted):

45, 55, 60, 65, 70, 75, 80, 85, 90, 95 (10 values)

- Median = average of middle two values:
 $P50 = (70 + 75) / 2 = 72.5$
-

Q2: Find the 70th percentile

Data (sorted):

20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70 (11 values)

- Position = $(70 / 100) \times (11 + 1) = 8.4$

- 8th = 55, 9th = 60
 - Interpolate:
 $P70 = 55 + 0.4 \times (60 - 55) = 57$
-

Q3: Percentile rank of 29

Data (sorted):

15, 18, 22, 24, 26, 29, 35, 37, 40, 42 (10 values)

→ 5 values are less than 29

Percentile Rank = $(5 / 10) \times 100 = 50$

So, 29 is at the 50th percentile.

Q4: Interpret percentile rank

You scored in the **92nd percentile** in a test of 5000 people:

People below you = $(92 / 100) \times 5000 = 4600$

✓ **Answer:** 4600 people scored less than you

Q5: Find Quartiles (25th, 50th, 75th percentile)

Data (sorted):

18, 20, 22, 23, 25, 27, 30, 32, 35, 38, 40, 42 (12 values)

- **Q1 = 25th percentile**
Position = $0.25 \times (12 + 1) = 3.25$
→ Between 3rd (22) and 4th (23):
 $Q1 = 22 + 0.25 \times (23 - 22) = 22.25$
- **Q2 = Median**
→ Avg of 6th and 7th = $(27 + 30) / 2 = 28.5$
- **Q3 = 75th percentile**
Position = $0.75 \times (12 + 1) = 9.75$
→ Between 9th (35) and 10th (38):

$$Q3 = 35 + 0.75 \times (38 - 35) = 37.25$$

Interpolation in Statistics

When calculating percentiles or missing values in a dataset, you might get a decimal position like:

Position = 2.4

This means the percentile lies between the 2nd and 3rd values.

Instead of rounding to the 2nd or 3rd value, we **interpolate**:

◆ Formula:

Let:

- x_i = value at position i
- x_{i+1} = value at position $i + 1$
- f = decimal part (for example, 0.4 from 2.4)

Then the **Interpolated Value** is:

$$\text{Interpolated Value} = x_i + f \times (x_{i+1} - x_i)$$

Example:

Sorted data: [10, 20, 30, 40, 50]

You want the value at position 2.6 → between 2nd (20) and 3rd (30) values.

Calculate:

$$\text{Interpolated Value} = 20 + 0.6 \times (30 - 20) = 20 + 6 = 26$$

So, you interpolated between 20 and 30 to get 26.

[part 2](#)

(Five number summary, covariance, correlation)